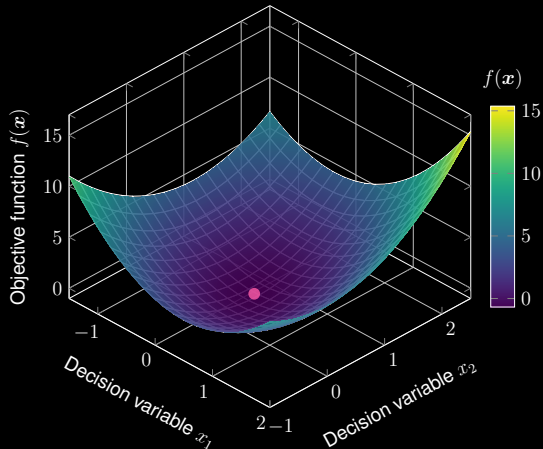
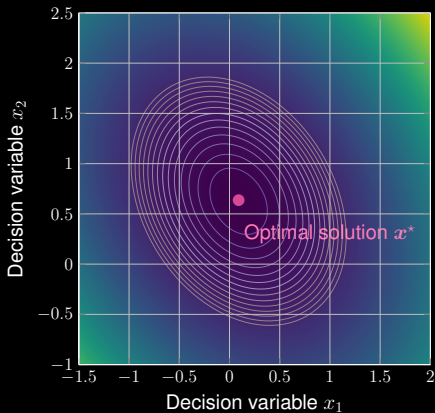


Quadratic Optimization

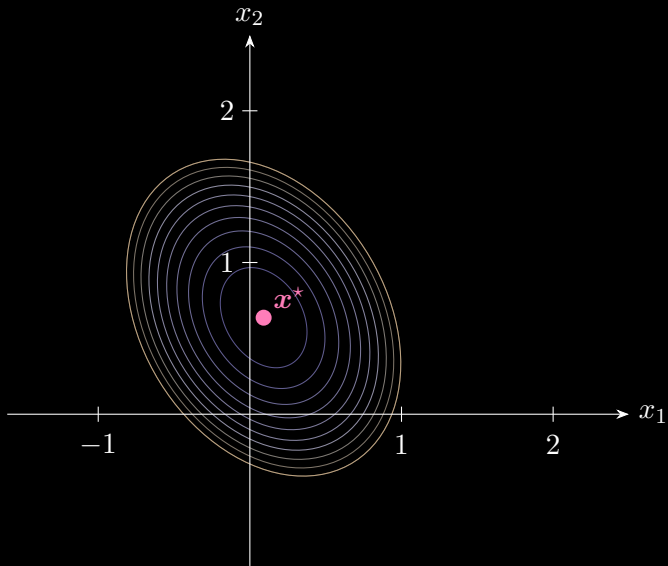
$$\min_{\mathbf{x} \in \mathbb{R}^2} \frac{1}{2} \mathbf{x}^\top \mathbf{A} \mathbf{x} - \mathbf{b}^\top \mathbf{x} \quad \text{with} \quad \mathbf{A} = \begin{bmatrix} 4 & 1 \\ 1 & 3 \end{bmatrix} \quad \text{and} \quad \mathbf{b} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$$



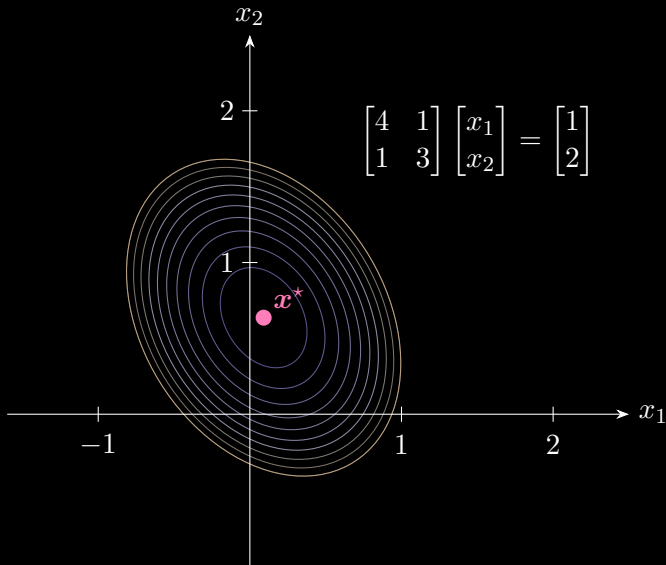
Quadratic Optimization

$$\min_{\mathbf{x} \in \mathbb{R}^2} \underbrace{\frac{1}{2} \mathbf{x}^\top \mathbf{A} \mathbf{x} - \mathbf{b}^\top \mathbf{x}}_{\text{Objective function } f} \quad \Rightarrow \quad \underbrace{\frac{d f}{d \mathbf{x}} = \mathbf{A} \mathbf{x} - \mathbf{b}}_{\text{First-order derivative}} = \mathbf{0} \quad \Rightarrow \quad \underbrace{\mathbf{A} \mathbf{x} = \mathbf{b}}_{\text{Linear system}}$$

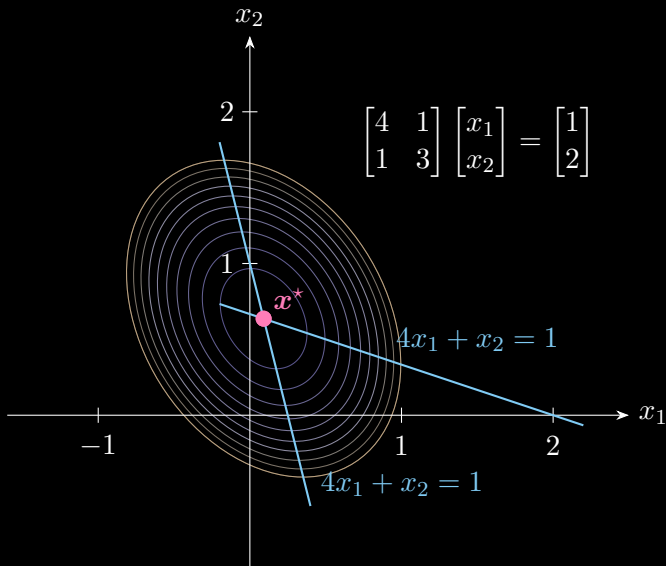
Quadratic Optimization



Equivalent Linear System $Ax = b$



Equivalent Linear System $Ax = b$



Update Equations

Essential idea: Conjugate gradient solves linear system $\mathbf{Ax} = \mathbf{b}$ iteratively.

❶ $\mathbf{x}$ is updated by

$$\mathbf{x}_{k+1} = \mathbf{x}_k + \underbrace{\alpha_k}_{\text{step size (unknown)}} \mathbf{d}_k$$

with initial points $\mathbf{x}_0$.

❷ $\mathbf{d}_k$ is the search direction:

$$\mathbf{d}_{k+1} = \underbrace{\mathbf{r}_{k+1}}_{\text{residual}} + \underbrace{\beta_k}_{\text{step size (unknown)}} \mathbf{d}_k$$

③ How to write down the update of residual $\mathbf{r}_{k+1}$?

$$\mathbf{x}_{k+1} = \mathbf{x}_k + \alpha_k \mathbf{d}_k$$

$$\Rightarrow \mathbf{Ax}_{k+1} = \mathbf{Ax}_k + \alpha_k \mathbf{Ad}_k$$

$$\Rightarrow \mathbf{b} - \mathbf{Ax}_{k+1} = \mathbf{b} - \mathbf{Ax}_k - \alpha_k \mathbf{Ad}_k$$

③ How to write down the update of residual $\mathbf{r}_{k+1}$?

$$\mathbf{x}_{k+1} = \mathbf{x}_k + \alpha_k \mathbf{d}_k$$

$$\Rightarrow \mathbf{A}\mathbf{x}_{k+1} = \mathbf{A}\mathbf{x}_k + \alpha_k \mathbf{A}\mathbf{d}_k$$

$$\Rightarrow \mathbf{b} - \mathbf{A}\mathbf{x}_{k+1} = \mathbf{b} - \mathbf{A}\mathbf{x}_k - \alpha_k \mathbf{A}\mathbf{d}_k$$

$$\Rightarrow \mathbf{r}_{k+1} = \mathbf{r}_k - \alpha_k \mathbf{A}\mathbf{d}_k$$

To summarize, we have the following update equations to solve $\mathbf{Ax} = \mathbf{b}$

$$\mathbf{x}_{k+1} = \mathbf{x}_k + \alpha_k \mathbf{d}_k \quad (\text{decision variables})$$

$$\mathbf{r}_{k+1} = \mathbf{r}_k - \alpha_k \mathbf{A} \mathbf{d}_k \quad (\text{residuals})$$

$$\mathbf{d}_{k+1} = \mathbf{r}_{k+1} + \beta_k \mathbf{d}_k \quad (\text{search direction})$$

△ But how to write down the update equations of α_k and β_k ?

Orthogonal residuals + Conjugate search directions ($\mathbf{A}$ -orthogonality)

$$\mathbf{r}_{k+1}^\top \mathbf{r}_k = 0 \quad \text{and} \quad \mathbf{d}_{k+1}^\top \mathbf{A} \mathbf{d}_k = 0$$

for any iterations k and $k + 1$.

④ Orthogonal Residuals $\mathbf{r}_k^\top \mathbf{r}_{k+1} = 0$:

$$\mathbf{r}_k^\top \underbrace{(\mathbf{r}_k - \alpha_k \mathbf{A} \mathbf{d}_k)}_{\mathbf{r}_{k+1}} = 0 \quad \Rightarrow \quad \alpha_k = \frac{\mathbf{r}_k^\top \mathbf{r}_k}{\mathbf{r}_k^\top \mathbf{A} \mathbf{d}_k}$$

④ Orthogonal Residuals $\mathbf{r}_k^\top \mathbf{r}_{k+1} = 0$:

$$\mathbf{r}_k^\top \underbrace{(\mathbf{r}_k - \alpha_k \mathbf{A} \mathbf{d}_k)}_{\mathbf{r}_{k+1}} = 0 \quad \Rightarrow \quad \alpha_k = \frac{\mathbf{r}_k^\top \mathbf{r}_k}{\mathbf{r}_k^\top \mathbf{A} \mathbf{d}_k}$$

○ Denominator:

$$\underbrace{\mathbf{r}_k^\top \mathbf{A} \mathbf{d}_k}_{\text{derived from the update equation of search direction}} = (\mathbf{d}_k - \beta_{k-1} \mathbf{d}_{k-1})^\top \mathbf{A} \mathbf{d}_k = \mathbf{d}_k^\top \mathbf{A} \mathbf{d}_k - \beta_{k-1} \underbrace{\mathbf{d}_{k-1}^\top \mathbf{A} \mathbf{d}_k}_{=0 \text{ (conjugate)}} = \mathbf{d}_k^\top \mathbf{A} \mathbf{d}_k$$

○ Update equation of step size α_k :

$$\alpha_k = \frac{\mathbf{r}_k^\top \mathbf{r}_k}{\mathbf{d}_k^\top \mathbf{A} \mathbf{d}_k}$$

⑤ Conjugate search directions $\mathbf{d}_{k+1}^\top \mathbf{A} \mathbf{d}_k = 0$:

$$\underbrace{(\mathbf{r}_{k+1} + \beta_k \mathbf{d}_k)}_{\mathbf{d}_{k+1}}^\top \mathbf{A} \mathbf{d}_k = 0 \quad \Rightarrow \quad \beta_k = -\frac{\mathbf{r}_{k+1}^\top \mathbf{A} \mathbf{d}_k}{\mathbf{d}_k^\top \mathbf{A} \mathbf{d}_k}$$

○ Considering

$$\mathbf{A} \mathbf{d}_k = \frac{\mathbf{r}_k - \mathbf{r}_{k+1}}{\alpha_k} \quad \alpha_k = \frac{\mathbf{r}_k^\top \mathbf{r}_k}{\mathbf{d}_k^\top \mathbf{A} \mathbf{d}_k}$$

○ Update equation of step size β_k :

$$\beta_k = \frac{\mathbf{r}_{k+1}^\top \mathbf{r}_{k+1}}{\mathbf{r}_k^\top \mathbf{r}_k}$$

Conjugate Gradient Method

To summarize, we have the following update equations to solve $\mathbf{Ax} = \mathbf{b}$

$$\alpha_k = \frac{\mathbf{r}_k^\top \mathbf{r}_k}{\mathbf{d}_k^\top \mathbf{A} \mathbf{d}_k} \quad (\text{step size})$$

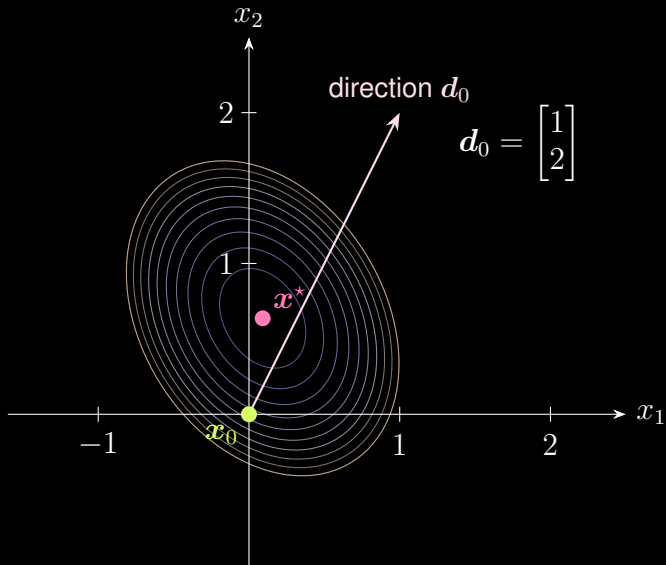
$$\mathbf{x}_{k+1} = \mathbf{x}_k + \alpha_k \mathbf{d}_k \quad (\text{decision variables})$$

$$\mathbf{r}_{k+1} = \mathbf{r}_k - \alpha_k \mathbf{A} \mathbf{d}_k \quad (\text{residuals})$$

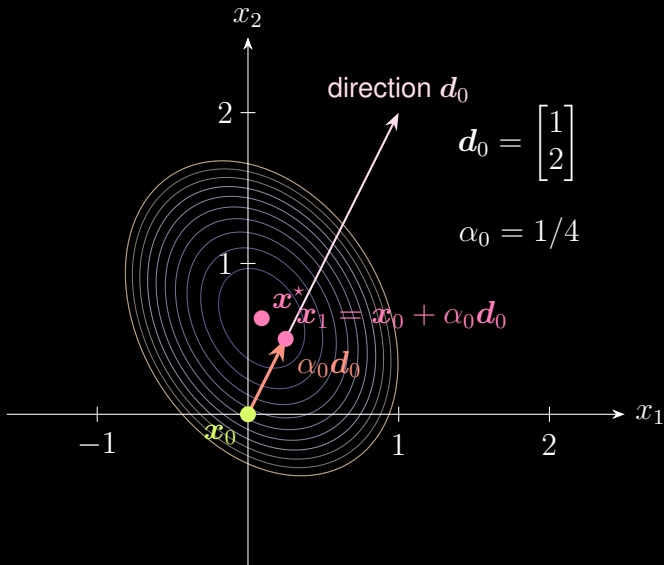
$$\beta_k = \frac{\mathbf{r}_{k+1}^\top \mathbf{r}_{k+1}}{\mathbf{r}_k^\top \mathbf{r}_k} \quad (\text{step size})$$

$$\mathbf{d}_{k+1} = \mathbf{r}_{k+1} + \beta_k \mathbf{d}_k \quad (\text{search direction})$$

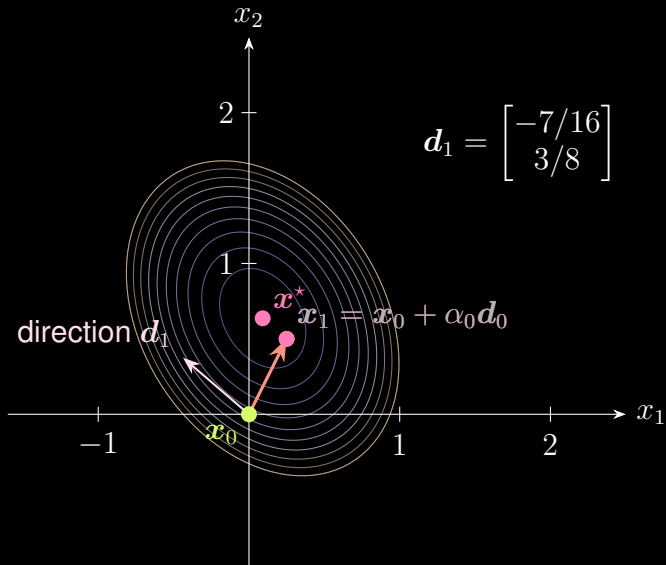
Conjugate Gradient Method



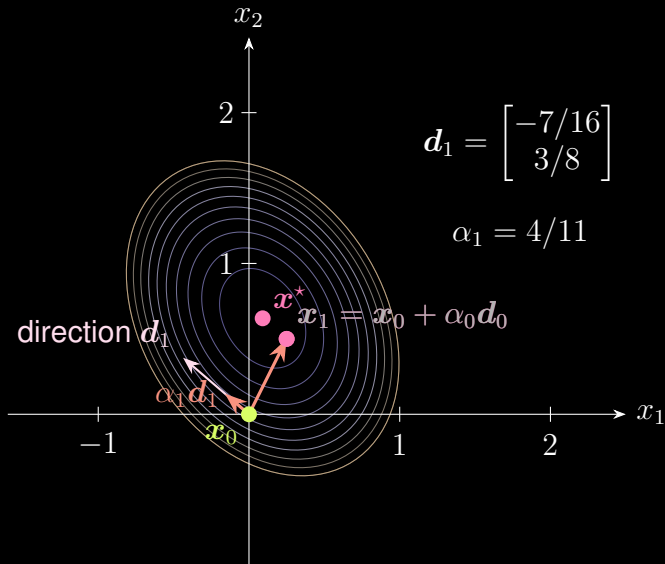
Conjugate Gradient Method



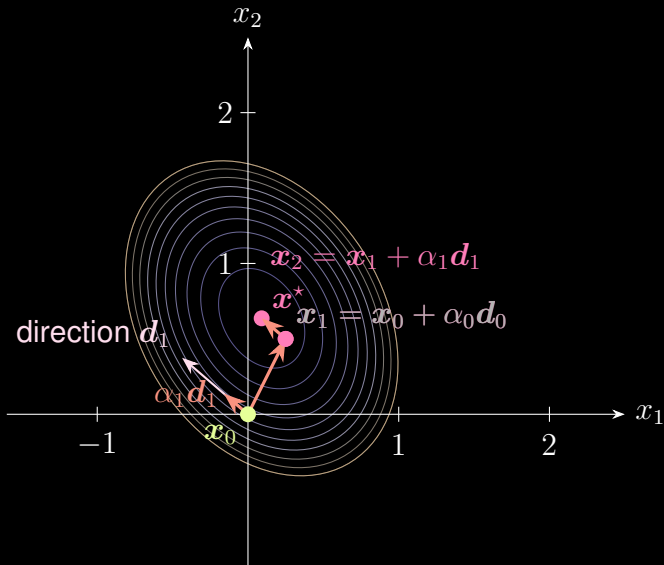
Conjugate Gradient Method



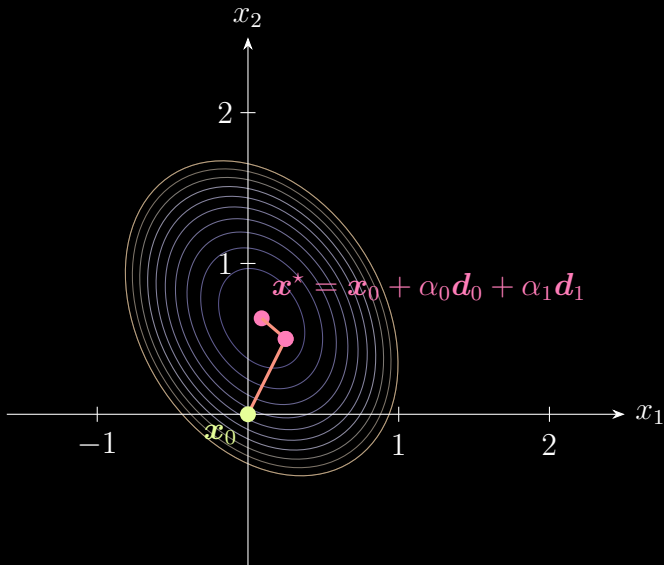
Conjugate Gradient Method



Conjugate Gradient Method



Conjugate Gradient Method



Thanks for your attention!

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